Received: 9 August 2023 | Revised: 16 January 2024 | Accepted: 7 March 2024 | CAAI Transactions on Intelligence Technology

DOI: 10.1049/cit2.12385

ORIGINAL RESEARCH

# An EfficientNet integrated ResNet deep network and explainable AI for breast lesion classification from ultrasound images

Kiran Jabeen[1] | Muhammad Attique Khan[1] | Ameer Hamza[1] | Hussain Mobarak Albarakati[2] | Shrooq Alsenan[3] | Usman Tariq[4] | Isaac Ofori[5]

[1]Department of Computer Science, HITEC University Taxila, Rawalpindi, Pakistan

[2]Computer and Network Engineering Department, College of Computing, Umm Al-Qura University, Makkah, Saudi Arabia

[3]Information Systems Department, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia

[4]Department of Management Information Systems, College of Business Administration, Prince Sattam bin Abdul Aziz University, Alkharaj, Saudi Arabia

[5]University of Mines and Technology, Tarkwa, Ghana

**Correspondence**

Isaac Ofori.
Email: iofori@umat.edu.gh

**Funding information**

Princess Nourah bint Abdulrahman University, Grant/Award Number: PNURSP2024R506

**Abstract**

Breast cancer is one of the major causes of deaths in women. However, the early diagnosis is important for screening and control the mortality rate. Thus for the diagnosis of breast cancer at the early stage, a computer-aided diagnosis system is highly required. Ultrasound is an important examination technique for breast cancer diagnosis due to its low cost. Recently, many learning-based techniques have been introduced to classify breast cancer using breast ultrasound imaging dataset (BUSI) datasets; however, the manual handling is not an easy process and time consuming. The authors propose an EfficientNet-integrated ResNet deep network and XAI-based framework for accurately classifying breast cancer (malignant and benign). In the initial step, data augmentation is performed to increase the number of training samples. For this purpose, three-pixel flip mathematical equations are introduced: horizontal, vertical, and 90°. Later, two pre-trained deep learning models were employed, skipped some layers, and fine-tuned. Both fine-tuned models are later trained using a deep transfer learning process and extracted features from the deeper layer. Explainable artificial intelligence-based analysed the performance of trained models. After that, a new feature selection technique is proposed based on the cuckoo search algorithm called cuckoo search controlled standard error mean. This technique selects the best features and fuses using a new parallel zero-padding maximum correlated coefficient features. In the end, the selection algorithm is applied again to the fused feature vector and classified using machine learning algorithms. The experimental process of the proposed framework is conducted on a publicly available BUSI and obtained 98.4% and 98% accuracy in two different experiments. Comparing the proposed framework is also conducted with recent techniques and shows improved accuracy. In addition, the proposed framework was executed less than the original deep learning models.

**KEYWORDS**

augmentation, breast cancer, classification, deep learning, optimization, ultrasound images

## 1 | INTRODUCTION

According to statistics, breast cancer is the fifth most common cancer death cause [1]. One of the primary causes of morbidity and mortality in women is breast cancer. Early detection and effective treatment of breast disease depend greatly on accurate diagnosis based on breast ultrasound (BUS) imaging [2]. The most accurate way to diagnose breast cancer is via a biopsy [3], and its pathology results are frequently used to diagnose. However, it necessitates more personnel and material resources and has the potential to harm a patient's natural physiological tissues, leading to both physical discomfort and psychological harm. Consequently, patients are not best served by a biopsy.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2024 The Author(s). *CAAI Transactions on Intelligence Technology* published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology and Chongqing University of Technology.

Medical imaging technology has evolved due to the quick advancements in computer technology, inspiring numerous novel techniques for diagnosing breast cancer [4]. Due to its adaptability, safety, and high sensitivity, ultrasonography (US) has been a common supplementary tool for detecting breast cancer in recent years [5]. Because ultrasound imaging has many benefits, such as no radiation, quicker imaging, higher sensitivity and accuracy, and lower cost, it has attracted increasing attention in recent years [6]. One of the least expensive imaging modalities in clinical medicine is breast ultrasound (BUS). A physical examination primarily seeks to identify breast tumours or other abnormalities. BUS is also risk-free and does not emit radiation [7]. Typically, breast ultrasound (BUS) images are of poor quality, have little contrast, and have a lot of uncertainty. These are the key root causes of the uncertainty: (i) The BUS photos are collected using various devices, at various times, and under various conditions; (ii) Different persons may have different breast traits; (iii) The BUS image has a low resolution, and speckles and noise are present by default. Additionally, the abilities and methods used by radiologists to obtain the images affect the quality of the BUS images [8].

Ultrasound is a method that depends on the operator. The diagnostic outcomes of doctors with various levels of medical training are significantly diverse, necessitating well-trained and experienced radiologists to evaluate ultrasound images [9]. Unfortunately, radiologists sometimes cannot correctly diagnose breast ultrasound images because of individual differences and low image quality. The computer-aided diagnosis (CAD) method has been developed to help provide more trustworthy and accurate diagnostic findings by reducing the inter-observer variation across various radiologists [6].

Image segmentation is typically a crucial step for further processing in a CAD system [10]. Therefore, creating a reliable and precise segmentation algorithm for BUS photos is crucial. However, because of the following problems, it is still difficult to distinguish lesions from BUS images: (i) Regardless of whether the lesions are benign or malignant, poor image quality causes blurry boundaries; (ii) The highly varying sizes, forms, and locations of the lesions present extra difficulties for segmentation algorithms [11]. The two main types of conventional BUS segmentation techniques are region-based and model-based. Using region-based approaches, image segmentation is done based on the similarity of nearby pixels or regions and some pre-existing restrictions, including the lesion regions' shape, appearance, and geographic placement. For building models employing distinctive prior/posterior-based energy functions, model-based techniques such as the graph-based, active contour, and Markov random-field models have several applications in BUS image segmentation [11].

### 1.1 | Problem statement, objectives, and major contributions

Deep learning-based classification techniques are frequently used to classify breast ultrasound pictures automatically, and researchers frequently suggest multi-image and multi-model fusion techniques [12]. Studies have demonstrated that it is possible to identify benign and malignant breast lesions utilising several BUI types [13]. BUS image analysis has been used in numerous studies to categorise breast cancers. It has been shown that morphological and textural characteristics are particularly helpful for distinguishing between benign and malignant tumours. Additionally, it has been claimed that integrating the two feature sets would increase the accuracy of tumour categorisation. The geometrical aspects of the tumour, such as its area, form, direction, regularity, and margins, are quantified by morphological features [14]. To address these challenges, we proposed a deep learning architecture based on the gated recurrent unit (GRU) module and a novel feature selection algorithm. Our major contributions of this work are defined as follows:

- Three-pixel flips mathematical equations are developed and performed data augmentation to improve the learning performance.
- EfficientNet-b0 and modified ResNet18 models are modified by adding GRU module and multi head self-attention. In addition, fine-tuning the last layers and trained using a deep transfer learning process that further analysed using Explainable AI (XAI) technique called GradCAM.
- A new feature selection technique is proposed based on the cuckoo search algorithm called cuckoo search controlled standard deviation and Newton Raphson.
- A novel fusion technique is proposed called parallel zero-padding max correlation coefficient.

## 2 | RELATED WORK

Numerous medical assistance systems have been created in recent years, and interest in computer-aided problem solving in the medical and health care industries is continually rising. A CBR approach is a common suggestion process approach in medical applications, particularly in medical aid systems [15]. When compared to other AI methods, CBR has several benefits. For example, it does not need an explicit domain model, and it requires less work to find important elements that accurately represent a situation. Additionally, a lot of data can be saved, and a CBR system can learn by picking up new information as instances [16]. Artificial intelligence (AI) tools are used in intelligent diagnosis processes to analyse patient data and medical information and generate potential diagnoses. AI systems can be used to quickly and accurately diagnose patients by effectively analysing large volumes of data and identifying patterns that a physician might overlook [17].

Breast Ultrasound image (BUI) analysis has been used in numerous studies to categorise breast cancers. It has been shown that morphological and textural characteristics are particularly helpful for distinguishing between malignant and benign tumours [18]. Convolutional neural networks (CNN) are used to classify and segment breast ultrasound images, which can be a novel and effective early diagnosis and analysis

method. Hossain et al. [19] presented a comparative analysis of vision transformers, ensemble and transfer learning models using explainable AI. The authors utilised a brain tumour dataset which consisted of 3264 images throughout the experiments and they achieved 94.5% highest accuracy and they evaluate the models using explainable AI. The limitation of this work was the less amount and less diversity of selected dataset. Daoud et al. [20] employed a pre-trained CNN model to correctly categorise BUS images, examining the use of main methods, such as the deep features extraction strategy and the transfer learning technique. The outcomes demonstrated that the transfer learning strategy falls short of the deep features extraction approach in performance. In addition, the results showed that effective feature selection and the extraction of deep features from the pre-trained CNN model have allowed for correct BUS image classification with an accuracy of 93.9%. Ding et al. [21] suggested an innovative MIL method for classifying BUS images. The instance space was first mapped to the idea space using a self-organising map. The bag feature vector is then built using the distribution of each bag instance within the concept space. A support vector machine (SVM) classifier was then used to categorise the tumours. The results of the experiments indicated that the suggested strategy could produce higher performance with an accuracy of 91.07%. Elsborg et al. [22] presented a framework based on LLM and explainable machine learning (ML) to analyse the biomarkers. The authors employed scBERT and regression algorithm. The authors used scRNA-seq dataset for the experimental work. Zhuang et al. [13] presented a trustworthy classification strategy for breast ultrasound images of cancers by fusing image decomposition and fusion methods with the adaptive spatial feature fusion method. The original breast ultrasound images were first processed using bilateral filtering and fuzzy enhancement methods. The original and mask photos were then used to create various breakdown images depicting the clinical traits of breast cancers. The best model was then chosen as the base model to extract deep learning features via transfer learning from a set of candidate deep learning models. Finally, the adaptive spatial feature fusion method was employed to train the network to achieve feature fusion and classification while training the classification network. A total of 1328 breast ultrasound photos were gathered for testing and training. The highest accuracy achieved by this method was 95.48%.

Singh et al. [23] suggested adding an atrous convolution layer to the conditional generative adversarial network segmentation algorithm for identifying tumour features at various resolutions of BUS images. As a result, the highest-level encoded feature respective relative effects were automatically rebalanced. Moreover, this work suggested incorporating a channel-wise weighting block into the network. The model was trained using the SSIM and L1-norm loss with the normal adversarial loss. The classification step demonstrated that a small number of statistical variables derived from the design of the anticipated mask's boundaries could accurately distinguish between malignant and benign tumours with an accuracy of 85%. Zhouet et al. [18] used the shearlet transform to extract textural feature descriptors. To assess the consistency, all these features were fed into two separate classifiers, support machine vector (SVM) and AdaBoost. According to the trial results, the classification accuracy for breast tumours was 91.0% by SVM and 90.0% by AdaBoost. The findings point to the possible use of the suggested technique for breast CAD in ultrasound images and its ability to classify the characteristics of ultrasound images of breast cancer. Yusufiyah et al. [24] suggested categorising the lesion's form type from a BUI using image processing. The Zernike and invariant moments were two different extraction feature approaches that were contrasted. The input image was filtered using the adaptive median filter and then segmented using the Chan-Vese active contour. Zernike moments and invariant moments were used for feature extraction, and SVM and multilayer perceptrons were used for classification. The suggested technique successfully classified 80% of the 45 photos. Shi et al. [25] enhance the representation performance of the original DPN by employing a stacked DPN (S-DPN) technique. The breast B-mode and prostate ultrasound elastography datasets were the two picture datasets used for tumour classification. Experimental outcomes showed that S-DPN achieved the best performance on breast and prostate ultrasound datasets, respectively, with 92.40% and 90.28% classification accuracies. Bing et al. [26] developed an innovative technique for classifying breast ultrasound images using multi-instance learning that was based on sparse representation. The global and local features were extracted using a concentric circle after image enhancement and segmentation to increase the accuracy of prediction and diagnosis. The problem of classification for ultrasound images was transformed into a MIL problem based on sparse representation. The representation of a bag was made up of a single feature vector created by combining all of the basis vectors in the dictionary into sparse linear combinations for each instance of the bag. The classification process combined the output of individual classifiers. About 84.72% of basic classifications were accurate and then employed LDA dimensional reduction once the initial BoW feature combinations were applied, four layers of concentric circles; then got classification accuracy of 86.34% in this way. Wang et al. [27] presented a survey on developing and implementation of Explainable methods for 6G technology. They concluded that the explainable method can overcome the risks by enhancing the transparency of the black-box AI decision-making process. Kalafi et al. [28] created a novel classification scheme for breast cancer lesions using a modified VGG16 architecture and an attention module. The chosen attention method improved the ultrasonic feature distinction between background and targeted lesions. To reduce the model difference between identified lesions and their labels, presented a new ensembled loss function that combined binary cross-entropy with the logarithm of the hyperbolic cosine loss. The network was optimised more quickly with this combined loss function. With an accuracy of 93%, the suggested model outperformed existing modified VGG16 architectures.

Eroğlu et al. [29] provided a hybrid-based CNN system to distinguish between normal, malignant, and benign breast lesions from US images. The Hybrid structure was built on the

Resnet50, Alexnet, and MobilenetV2 models. These model's features were acquired and concatenated independently. As a result, more features were being used. Then, using ML classifiers like SVM and KNN, the most useful features were chosen using the mRMR (Minimum Redundancy Maximum Relevance) feature selection technique. The SVM classifier achieved the highest rate with an accuracy of 95.6%. Silva et al. [30] presented an architecture for data augmentation using an explainable authorship. The authors used LLM models for designing dataset augmentation techniques. The utilised dataset consisted of 1000 samples and they tested to generate an abstract from the multimodal transformer. Finally, moon et al. [31] presented a CAD system that detected tumours utilising an image fusion approach. Various image content representations and an ensemble of various CNN architectures on US images were employed in this approach. The suggested CNN-based approach consisted of DenseNet, ResNet, and VGGNet. There were 697 tumours in the open dataset (BUSI), comprising 133 normal pictures, 210 malignant tumours, and 437 benign lesions. The obtained accuracy of this method was 94.62%.

In summary, the above techniques focused on the following points: (i) The techniques mentioned above did not focus on the augmentation process (imbalanced dataset) that later reduced the accuracy; (ii) Deep learning models that they used are not fine-tuned in terms of skipping/adding layers, and (iii) single source features are not enough for the target accuracy, but some of the researchers tried to resolve this issue but missed the challenge of redundant and irrelevant features. In this work, we proposed a new data-augmented deep learning and optimal selection of fused features to accurately classify breast cancer from BUSI.

## 3 | MATERIALS AND METHODS

In this section, a deep learning pipeline is proposed for the classification of breast ultrasound images. The proposed framework is presented in Figure 1. This Figure illustrates that BUSI dataset is utilised for the experimental process. Initially, the data augmentation is implemented on the selected dataset to defeat the imbalance problem. After that, two modified models are employed, named ResNet-18 with self-attention and GRU-based EfficientNet-b0. Both models are trained using deep transfer learning. Self-attention and GRU activations are utilised to extract the features. Following that, the cuckoo search controlled standard deviation and Newton Raphson-based feature selection technique is employed on extracted feature in order to select the best features. In the next step, a minimum correlated serially fused approach is developed for the fusion of best features that are later passed to the feature selection step (second time). The final classification is performed in the last phase using multiple classifiers.

### 3.1 | Dataset

Breast ultrasound images of women between 25 and 75 are included in the baseline data. This information was gathered in 2018 [31]. There were 600 female patients in all. The dataset comprises seven hundred eighty photos with an average size of 500 by 500 pixels. The pictures are PNG files. The photos are grouped into three categories: normal, benign, and malignant. Figure 2 shows the sample images of the selected dataset.

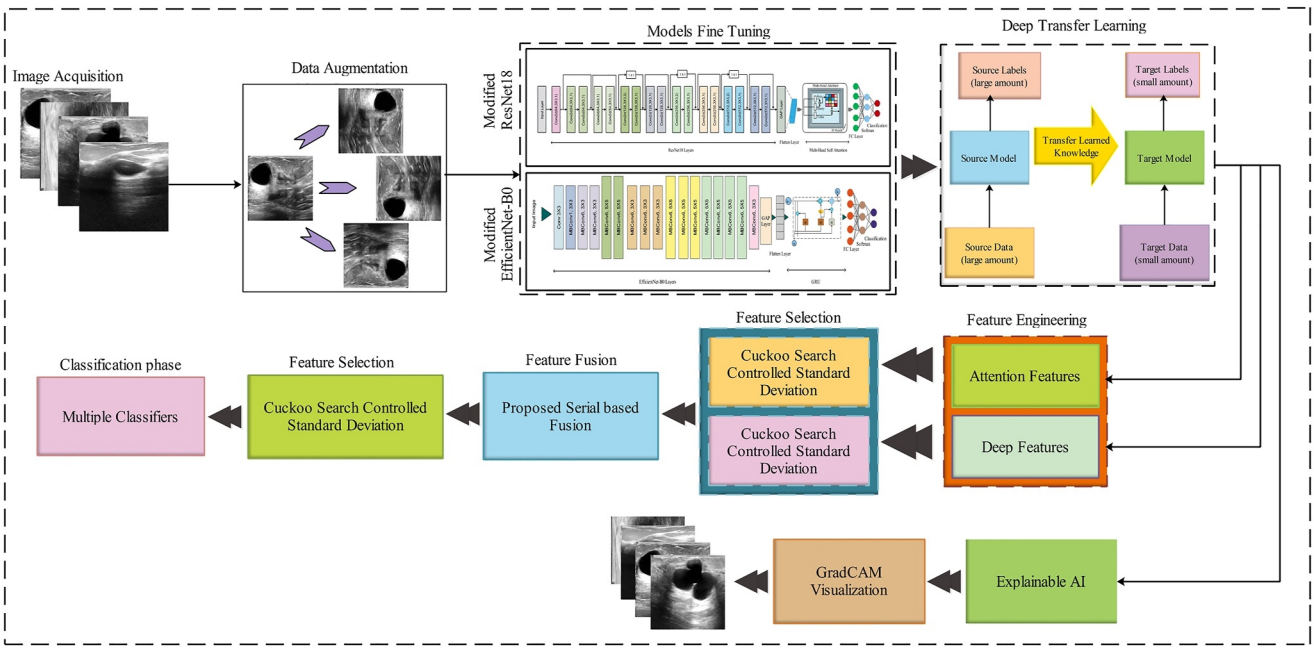


**FIGURE 1** The proposed augmented deep learning and optimal fused features framework of breast cancer classification.

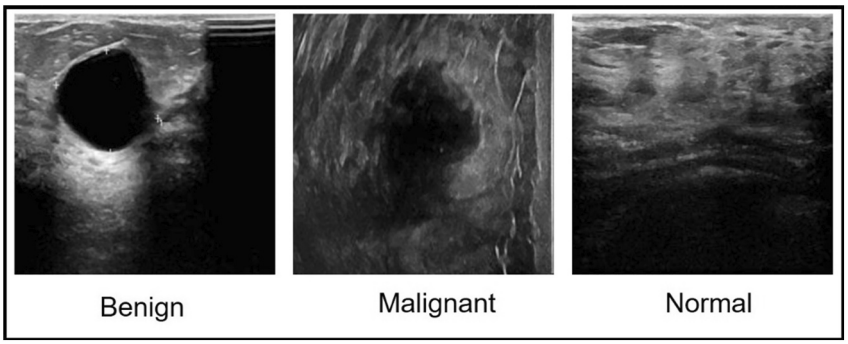


**FIGURE 2** Sample ultrasound images of the BUSI dataset. BUSI, breast ultrasound imaging dataset.

## 3.2 | Data augmentation

Various methods, collectively called 'data augmentation', can artificially increase the amount of data by producing more data points from current data. Data augmentation enhances the ML model's performance and outcome by adding unique or distinctive examples in the training datasets [32]. As a result, large and sufficient datasets improve the performance and accuracy of ML models (deep learning). In this work, three mathematical equations are implemented for the augmentation process to expand the diversity of the original dataset. These equations are defined as follows:

The original image is denoted by $\Psi(x, y)$, and $\Delta$ denotes the entire dataset $\Delta \in \Psi(x, y)$. The first equation is implemented to shift all image pixel to right side. In the second operation, all image pixels of the original image pixels are in vertical order. Finally, in the third equation, all image pixels of the equation are in vertical order.

$$Y1(x, y) = \sqrt{\Psi(x, y)} \text{ and } \Delta \in \Psi(x, y) \tag{1}$$

$$Y2(x, y) = -\sqrt{\Psi(x, y)} \text{ and } \Delta \in \Psi(x, y) \tag{2}$$

$$Y3(x, y) = \sqrt{Y1(x, y)} \text{ and } \Delta \in Y1(x, y) \tag{3}$$

The sample outputs of these above equations are shown in Figure 3. After augmentation, the total target number of images is approximately 12,000. Therefore, each class has above than 3000 images.

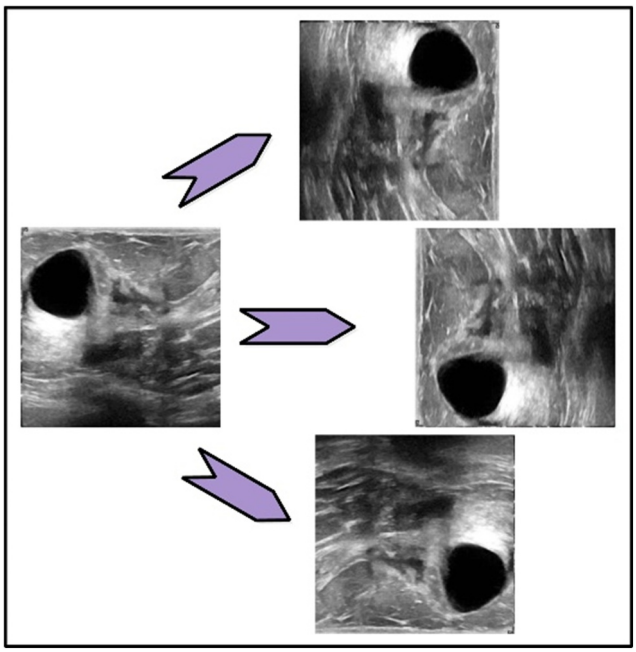

**FIGURE 3** Visual sample images of data augmentation using three mathematical equations.

## 3.3 | EfficientNet-b0 integrated ReNet model

**Modified EfficientNet-b0:** Using more than a million photos from the ImageNet collection, the convolutional neural network EfficientNet-b0 was trained. The pre-trained network can identify photos into 1000 object categories, including several things, a tiger, a banana, and a mobile. The network has therefore acquired rich image features for a variety of images. In addition, the network handles images with a resolution of 224 by 224. In this research, the EfficientNet-b0 is modified using GRU module. The last three layers including fully connected, softmax and classification are removed. After global average pool layer, a flatten layer is added in order to convert the 2D information into 1D. After that, GRU is added with 1280 hidden units, state activation function is tanh and gate activation function is log-sigmoid. In the end, FC layer, softmax and classification is attached and fine-tuned using deep transfer learning. By this modification, the model has ability to acquire hierarchical features from images and encode consecutive connections leading to an extensive and insightful representation of the input images. The deep feature representation enhanced specific to the task efficiency. The modified efficient-b0 architecture presented in Figure 4.

**Modified ResNet-18** is an 18-layer convolutional neural network. The ImageNet database contains a pretrained version of the network trained on more than a million photos. Several things, like a car, a monkey, and a pen, are just a few of the 1000 object categories into which the pretrained network can classify images. As a result, the network has accumulated rich visual information for a range of images. The network accepts images with a resolution of 224 by 224. In this work, we proposed a modified ResNet-18 for the classification task. The model is modified by adding the multi-head self-attention mechanism in ResNet-18. The last three layers are removed and a flatten layer is added. After flatten layer, a multi-headed self-attention is added with six heads and 384 keys. By this updation, the model captures detailed contextual information through exploring the associations between all points in the input feature maps. This adaptability makes the model suitable for tasks with erratic patterns. The modified model is shown in Figure 5.

EfficientNet-B0 selected for this work due to its cutting edge performance, capacity for transfer learning, and efficiency with respect to parameters. On the other hand, ResNet-18 came out on top due to its proven performance on a range of datasets, interpretability, ease of use, and faster training time.

## 3.4 | Transfer learning and deep features extraction

Transfer learning seeks to enhance the performance of target learners on target domains by transferring the knowledge found in multiple but related source domains [33]. It uses a learning system that it has learnt from a dataset to tackle a new, related problem more rapidly and efficiently. Figure 6 illustrated the working of transfer learning.

Two components comprise a domain $\partial$: a feature space $V$ and a marginal distribution $f(V)$. In simple words $\partial = \{V, f(V)\}$. Additionally, the symbol $V$ stands for an instance set, defined as $V = \{V | V_l \in V, l = 1,2,3\ldots m\}$. A task $\tau$ is made up of a decision function $\propto$ and a label space $\rho$, so $\tau = \{\rho, \propto\}$. It is expected that the decision function $\propto$ is an implicit one for being discovered from the sample data. Several hyper-parameters have been initialised in the TL process, such as learning rate, momentum, regularisation factor, epochs, and optimiser. The learning rate of the ResNet model is selected as 0.0004 and 0.0002 for EfficientNet-b0. These values are computed after the number of hit and trail-based approaches. The stochastic gradient descent (SGD) optimiser is employed for both fine-tuned models. The number of epochs for both deep models has been initialised with 100 and 0.743 momentum values.

After training both fine-tuned models, we activated the global average pooling layers of both models and extracted deep features. The deep testing features are obtained in the form of feature vectors such as $F_{\text{res}} = N \times 512$ and $F_{\text{eff}} = N \times 1280$.

## 3.5 | Feature selection

Feature selection, known as a data pre-processing strategy, has shown significant improvement in ML for preparing high-dimensional data [34]. The purpose of feature selection is to make the input more useful in terms of redundant and irrelevant information removal for accurate classification purposes.

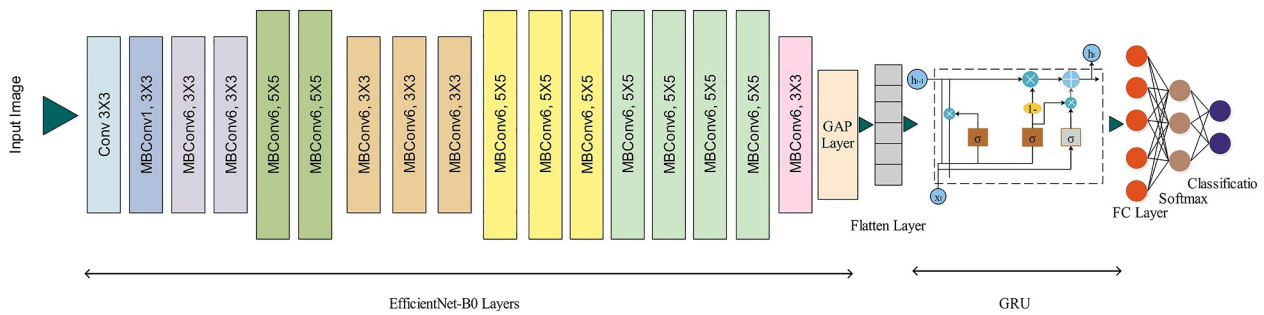


**FIGURE 4** Modified with GRU module architecture of EfficientNet-b0. GRU, gated recurrent unit.

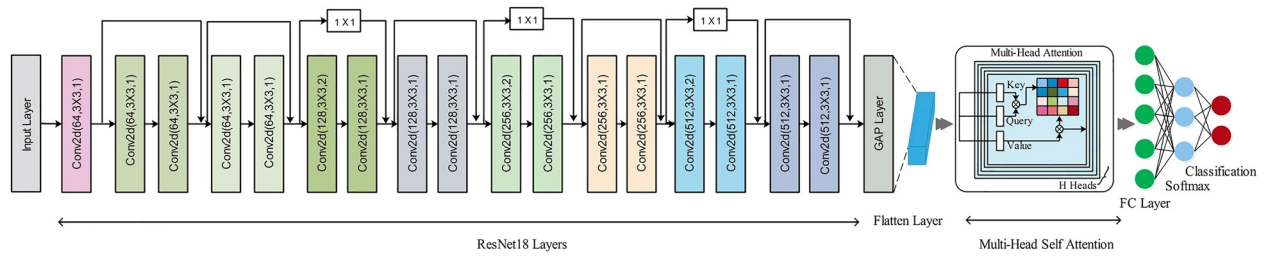


**FIGURE 5** Modified architecture of ResNet-18.

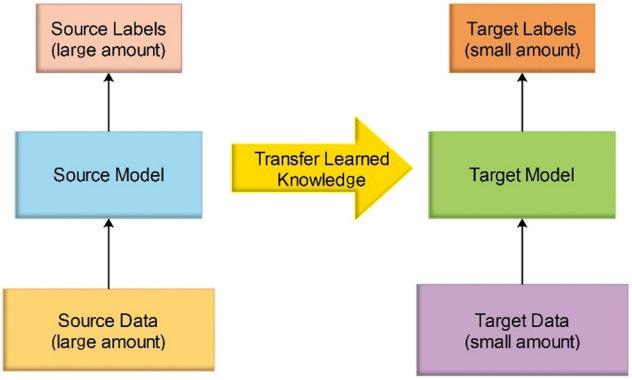


**FIGURE 6** The process of transfer learning.

However, the high-dimensional data not only increase the computational time but is also involved in the problem of overfitting [35].

In this work, we proposed a new feature selection technique based on the cuckoo search algorithm and Newton Raphson method. The proposed technique called cuckoo search controlled standard deviation and Newton Raphson. In this technique, initially, features are selected using the cuckoo search algorithm, and then standard deviation value-based features are passed to the next phase. Finally, the stopping criteria are defined based on the new function that is based on the Newton-Raphson approach. The cuckoo sear technique is initially stary with the cuckoo breeding behaviour. As, the **c**uckoos have a more active reproductive strategy than other bird species; hence, sometimes they remove the eggs of other cuckoo bird species to enhance the likelihood that their eggs will hatch. The cuckoo throws the egg away or departs the nest entirely before moving to make a new nest elsewhere. After this step, levy flight step is opted. The Lévy flights is based on the random walks with random directions. Compared to a conventional random walk, Lévy flights can cut the number of optimisation algorithm rounds by around four orders [34].

**Cuckoo Search Algorithm:** The following three rules form the basis of the CS optimisation algorithm: (i) One egg is incubated in a nest randomly selected by each cuckoo; (ii) best nests that produce the finest quality eggs will be passed down to the following generation; (iii) for a particular number of nests, a host cuckoo has a chance of [0,1] of discovering a foreign egg. In this scenario, the host cuckoo has two choices: toss the egg or leave the nest and make a new one somewhere.

A fraction may also represent this final assumption $z_o$ of the $n$ host nests that are changed out for new nests. A maximising problem of solution efficiency or suitability can simply be proportional to the objective function's importance. According to how the fitness function in the evolutionary algorithm is defined, other kinds of fitness can also be expressed. For implementation, we may use the following straightforward representations. Each egg in a nest denotes a solution, and each cuckoo can only lay one egg. The goal is to utilise new and potential intentions to gain (cuckoos) to remove and replace subpar ones in the nests. It is noted that this technique may be improved in the more complex circumstance when each nest has numerous eggs signifying a variety of solutions. Let's say there is only one egg in each nest. Because each nest stands in for one egg, which likewise serves as a stand-in for one cuckoo, there is no difference among the cuckoo, nest, or egg. A switching parameter called $z_o$ balances the usage of a local random walk and a global explorative random walk in this technique. We can write the local random walk as follows [34]:

$$v_n^{\tau+1} = v_n^{\tau} + \partial c \otimes R(z_o - \varepsilon) \otimes \left(v_m^{\tau} - v_l^{\tau}\right) \tag{4}$$

where $R$ represents the Heaviside function, $\partial$ denotes a random number chosen at random from a uniform distribution, $c$ is the step size, and $v_m^{\tau}$ and $v_l^{\tau}$ are two distinct solutions chosen randomly by random permutation. On the other side, Lévy flights are used to influence the global random walk [34].

$$v_n^{\tau+1} = v_n^{\tau} + \partial k(c, \mu) \tag{5}$$

$$k(c, \mu) = \frac{\mu\Gamma(\mu)\sin(\pi\mu/2)}{\pi} \frac{1}{c^{1+\mu}}, (c \gg c_0 > 0) \tag{6}$$

The step size scaling factor, in this case, must be connected to the scales of the relevant problem, $\partial > 0$. Sometimes, we can choose $\partial = O(k/10)$, where $k$ is the relevant problem's characteristic scale, but occasionally $\partial = O(k/100)$ can be more efficient and prevent us from going too far. The equation described above is the stochastic equation for a random walk. A random walk is a Markov chain that generally is based on its present location and the transition probability to determine its future state or location (the second term). However, a sizeable portion of the new solutions should be produced through far-field randomisation to prevent the system from becoming stuck in a local optimum. Their locations must be sufficiently removed from the recent best solution.

## 4 | NOVELTY: STANDARD ERROR MEAN COMPUTATION FUNCTION

A new sub-step selection function is proposed in this work to improve the performance of cuckoo search algorithm. The new function is based on the standard deviation (SD) and standard error means (SEM). The SD measure the amount of variability of the selected features of cuckoo search algorithm. Based on the SD, the SEM is computed that measure how far the mean value is likely to be from selected features in the previous stage. First, the $\sigma$ is computed as follows by employing these selected features.

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}\left(f_i - \bar{f}\right)^2}{n-1}} \tag{7}$$

where $f_i$ denotes the current feature from the selected vector of cuckoo search, $\overline{f}$ denotes the mean value of the entire selected vector, and $n$ denotes the total sample images, respectively. After that, the standard error is computed using $\sigma$ value as follows:

$$\sigma_{\overline{x}} = \frac{\sigma}{\sqrt{n}} \tag{8}$$

Based on the value of $\sigma_{\overline{x}}$, the following function is defined as follows:

$$\text{Funct} = \begin{cases} \text{Sel}(k) & \text{for } f_i \geq \sigma_{\overline{x}} \\ \text{Ignore}, & \text{Elsewhere} \end{cases} \tag{9}$$

In the next step, the probability is computed of Sel($k$) and consider the most highest appeared feature value. The most appeared feature is added in the defined function for final selection.

$$\text{Prob} = \frac{n(\text{Sel}(k))}{\text{Total Features}} \tag{10}$$

$$\text{Maximum} = \arg\max_{k=1}[\text{Prob}(\text{Sel}(k))] \tag{11}$$

$$\text{Fnsl} = \begin{cases} \text{Fsl}(k) & \text{for Sel}(k) \geq \text{Maximum} \\ \text{Ignore}, & \text{Elsewhere} \end{cases} \tag{12}$$

where $Fsl(k)$ denotes the final selected features of each iteration. The fitness is calculated after each iteration. The Fine-KNN is employed a fitness function. The extracted features are passed to the proposed cuckoo search controlled standard deviation and Newton Raphson feature selection. The resultant feature dimensions are $N \times 261$ and $N \times 629$, respectively. Where $N$ denotes the total number of observations. The selected features are employed to a newly proposed feature fusion.

## 4.1 | Novelty: Feature fusion

Fusion is the process of integrating distinct extracted characteristics that are kept in a dictionary to create a single feature file that is extremely informative. Different metrics must be tailored to the application in order to meet various user needs, according to the relative analysis of feature fusion approaches. In this work, we proposed a zero-padding parallel correlation coefficient approach for feature fusion. Consider, we have two optimal feature vectors $\phi f_1$ and $\phi f_2$, where dimensions of each vector is $N \times 261$ and $N \times 629$, respectively. First, the zero-padding is performed for the purpose of equal dimensions of both vectors. The maximum feature vector dimension is selected in this work; therefore, the target length is $N \times 629$. The zero-padding is defined as follows :

$$Pd = \text{padding}(\phi f_1, 0) \tag{13}$$

After that, the similarity is computed among both vectors base on the following formulation:

$$r(\phi f_1, \phi f_2) = \frac{\sum_M \sum_N \left(f_1 - \overline{f_1}\right)\left(f_2 - \overline{f_2}\right)}{\sqrt{\left(\sum_M \sum_N \left(f_1 - \overline{f_1}\right)^2\right)\left(\sum_M \sum_N \left(f_1 - \overline{f_1}\right)^2\right)}} \tag{14}$$

where $M$ and $N$ have the same dimension of both feature vectors (629 in this work). The value of $r$ should be in the range of [−1, 1]. Hence, the absolute value is computed for $r$ to quantify the similarity between $f_1$ and $f_2$.

$$r(\phi f_1, \phi f_2) = \left| \frac{\sum_M \sum_N \left(f_1 - \overline{f_1}\right)\left(f_2 - \overline{f_2}\right)}{\sqrt{\left(\sum_M \sum_N \left(f_1 - \overline{f_1}\right)^2\right)\left(\sum_M \sum_N \left(f_1 - \overline{f_1}\right)^2\right)}} \right| \tag{15}$$

We consider $r \leq 0.30$ in this work as a minimum similarity among features. Hence, we fused only the minimum similarity features. The final fusion function is defined as follows:

$$\text{Fus}(\phi f_1, \phi f_2) = \begin{cases} Fus(k) & for\ s(\phi f_1, \phi f_2) \leq r \\ \text{Max Similarity} & \text{Ignore Features} \end{cases} \tag{16}$$

where $Fus(k)$ denotes the final fused feature vector of size $N \times 782$. To optimise this fused vector, the proposed optimisation algorithm is applied and selected the best features of vector size $N \times 466$. This resultant vector is passed to the ML classifiers for the final classification.

**Algorithm Proposed Breast Cancer Classification using Deep Learning**

```
Input: Original Ultrasound Images
Output: Predicted Label using Proposed Framework
Step 1: Acquisition of original images and performed data augmentation using Equations (1-3).
Step 2: Fine-tuned deep learning model ResNet-18 and EfficientNet-b0.
Step 3: Train model using deep transfer learning
Step 4: Select the optimal features using proposed optimisation technique
Step 5: Objective function f(v), v = (v1, v2, v3...vm)
Step 6: Create a baseline population of m host nests
```

```
While (stop criteria)
Select a cuckoo at random (let's say, l) and
replace its solution with Lévy flights
Analyse its fitness and quality f_l
[ for maximisation, f_l ∝ f(v_l)]
Pick one nest at random from n (let's say, m)
if (f_l > f_m)
Replace m with the new solution
end if
The worst nests are abandoned to some extent
z_o, and new ones are constructed
keep the top or best nests or solutions
Find the current best by ranking the
solutions/nests
provide the upcoming generation with the
best options available now
end while
Step 7: Find the SERM of the best solutions
and computed the fitness
Step 8: Returned two best feature vectors
ϕf_1,ϕf_2
Step 9: Fused features using proposed fusion
approach
Step 10: Again Repeat Step 4-8 and obtained a
single best selected feature vector
Step 11: Performed classification using
machine learning classifiers.
```

## 5 | RESULTS AND DISCUSSION

**Dataset and Performance Measures:** The proposed framework is evaluated on a publicly available ultrasound breast cancer imaging dataset comprising three classes- normal, benign, and malignant. The detail of the dataset is given in Section 3. Several ML classifiers have been opted for, such as SVM, Neural network, and named a few more. The performance of each classifier is computed based on the accuracy measure (Acc), sensitivity rate (SER), precision rate (PREC), F1-Score (F1S), AUC, FNR, and computational time (seconds).

**Experimental Setup:** For the training of the deep models, we divide the dataset into a ratio of 50:50. This represents that 50% of the images of each class have been utilised for the training of the models and the remaining 50% used for the testing. The 10-fold cross-validation was performed during the training and testing process. In addition, several hyperparameters have been used to train deep models. The used hyperparameters are learning rate, epochs, momentum, minibatch size, and optimiser. The values of these hyperparameters are 0.0001, 50, 0.7, 32, and SGD. The entire experimental process was conducted on MATLAB2022a using a personal computer Corei7 having 16GB of RAM.

**Numerical Results:** The numerical results of the proposed framework have been presented in this section. The detailed six experiments have been performed to evaluate the entire framework. In the first experiment, fine-tuned ResNet-18 deep model features are extracted and performed classification. Table 1 presents the numerical results of this experiment. In this experiment, the Ensembled Subspace KNN (ESKNN) classifier obtained the highest accuracy of 98.3%. The SER of this classifier is 98.36%, the PREC is 98.33%, F1S is 98.34%, and FNR is 1.64%. These obtained measures can be verified by a confusion matrix as illustrated in Figure 7. The diagonal values in this figure show the true prediction rate. The second-best accuracy is obtained by Fine KNN classifier of 98.2%. The minimum obtained accuracy for this experiment is 95.9% by Tri-layered NN. The computational time of each classifier is also noted, and it is observed that the medium NN has a minimum time of 8.3131 (sec).

In the second experiment, fine-tuned EfficientNet-b0 deep model features are extracted and performed classification. Table 2 presents the numerical results of this experiment. In this table, it is noted that the ESKNN classifier obtained the highest accuracy of 97.6%. The SER of this classifier is 97.63%, PREC is 97.6%, F1S is 97.61%, and FNR is 2.37%. These measures can be verified by a confusion matrix, as illustrated in Figure 8. From

**TABLE 1** Classification results of fine-tuned deep ResNet-18 features on augmented BUSI dataset.

| Classifier | SER (%) | PREC (%) | F1S | AUC | Acc (%) | FNR | Time (sec) |
|---|---|---|---|---|---|---|---|
| ESKNN | 98.36 | 98.33 | 98.34 | 0.99 | **98.3** | 1.64 | 22.08 |
| Fine KNN | 98.2 | 98.2 | 98.2 | 0.98 | 98.2 | 1.8 | 21.139 |
| CSVM | 97.5 | 97.4 | 97.44 | 1 | 97.4 | 2.5 | 16.49 |
| MGSVM | 97.36 | 97.3 | 97.32 | 1 | 97.3 | 2.64 | 21.588 |
| Weighted KNN | 97.06 | 96.93 | 96.99 | 1 | 97 | 2.94 | 18.697 |
| $WN^2$ | 96.86 | 96.86 | 96.86 | 0.99 | 96.8 | 3.14 | 11.443 |
| QSVM | 96.8 | 96.73 | 96.76 | 0.99 | 96.7 | 3.2 | 15.459 |
| $MN^2$ | 96.46 | 96.43 | 96.44 | 0.99 | 96.4 | 3.54 | **8.3131** |
| $BiN^2$ | 96.46 | 96.43 | 96.44 | 0.98 | 96.4 | 3.54 | 22.556 |
| $TriN^2$ | 96 | 95.96 | 95.98 | 0.98 | 95.9 | 4 | 24.296 |

*Note*: Bold denotes the best results.

Abbreviations: BiN2, bilayered neural network; CSVM, cubic SVM; ESKNN, ensembled subspace KNN; MGSVM, medium Gaussian SVM; MN2, medium neural network; TriN2, trilayered neural network; WN2, wide neural network.

**TABLE 2** Classification results of fine-tuned deep EfficientNet-b0 features on augmented BUSI dataset.

| Classifier | SER (%) | PREC (%) | F1S | AUC | Acc (%) | FNR | Time (sec) |
|---|---|---|---|---|---|---|---|
| ESKNN | 97.63 | 97.6 | 97.61 | 0.99 | **97.6** | 2.37 | 61.76 |
| Fine KNN | 97.26 | 97.2 | 97.22 | 0.98 | 97.2 | 2.74 | 56.674 |
| CSVM | 96.36 | 96.3 | 96.32 | 0.99 | 96.3 | 3.64 | 81.763 |
| MGSVM | 95.5 | 96.3 | 95.89 | 0.99 | 95.5 | 4.5 | 121.85 |
| Weighted KNN | 94.36 | 94.33 | 94.34 | 0.99 | 94.2 | 5.64 | 58.614 |
| $WN^2$ | 94.73 | 94.66 | 94.69 | 0.99 | 94.6 | 5.27 | 19.212 |
| QSVM | 95.06 | 95 | 95.02 | 0.99 | 95 | 4.94 | 69.952 |
| $MN^2$ | 94.2 | 94.13 | 94.16 | 0.98 | 94.1 | 5.8 | **14.621** |
| $BiN^2$ | 92.7 | 92.66 | 92.68 | 0.96 | 92.6 | 7.3 | 42.235 |
| $TriN^2$ | 92.7 | 92.63 | 92.66 | 0.96 | 92.6 | 7.37 | 68.024 |

*Note*: Bold denotes the best results.

Abbreviations: BiN2, bilayered neural network; CSVM, cubic SVM; ESKNN, ensembled subspace KNN; MGSVM, medium Gaussian SVM; MN2, medium neural network; TriN2, trilayered neural network; WN2, wide neural network.

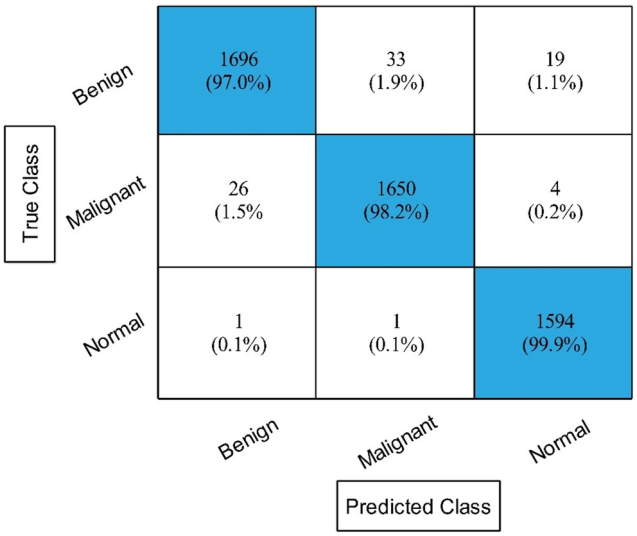


**FIGURE 7** The confusion matrix of experiment 1 for ESKNN classifier. ESKNN, Ensembled Subspace KNN.

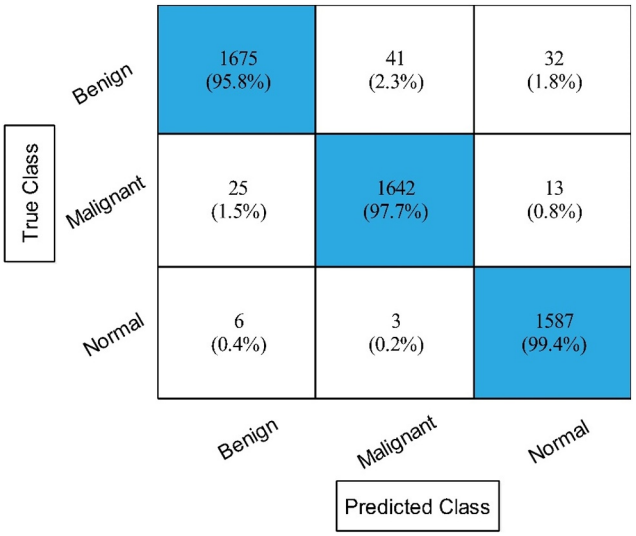


**FIGURE 8** The confusion matrix of experiment 2 using ESKNN classifier. ESKNN, Ensembled Subspace KNN.

this figure, the diagonal values show the true prediction rate. The second best accuracy is obtained by Fine KNN classifier of 97.2%. The minimum obtained accuracy for this experiment is 92.6% by Tri-layered NN. Computational time of each classifier is also noted and it is observed that the medium NN has the minimum time of 14.621 (sec). After this experiment, it is observed that the accuracy is reduced and computational time is increased as compared to previous experiment.

In the third experiment, the improved cuckoo search optimiser is applied on the ResNet19 deep features and selects the best of them for the classification. Table 3 presents the numerical results of this experiment. In this table, it is noted that the ESKNN classifier obtained the highest accuracy of 98%. The SER of this classifier is 98.06%, PREC is 98.06%, F1-Score is 98.06%, and FNR is 1.94%. These measures can be verified by a confusion matrix as illustrated in Figure 9. From this figure, the diagonal values show the true prediction rate. The second best accuracy is obtained by Fine KNN classifier of 97.9%. The minimum obtained accuracy for this experiment is 95% by Tri-layered NN. Computational time of each classifier is also noted and it is observed that the medium NN has the minimum time of 4.1722 (sec). In this experiment, the accuracy is increased and computational time is decreased comparatively previous experiment.

In the fourth experiment, the improved cuckoo search optimiser is utilised to select the best features of the EfficientNet-b0 model and later employed for the classification. Table 4 presents the numerical results of this experiment. In this table, it is noted that the ESKNN classifier obtained the highest accuracy of 97.5%. The SER of this classifier is 97.5%, the PREC is 97.43%, F1-Score is 97.46%, and FNR is 2.5%. A confusion matrix can verify these measures, as illustrated in

**TABLE 3** Classification results of Resnet18 deep model after employing Cuckoo Search based best feature selection on augmented BUSI dataset.

| Classifier | SER (%) | PREC (%) | F1S | AUC | Acc (%) | FNR | Time (sec) |
|---|---|---|---|---|---|---|---|
| ESKNN | 98.06 | 98.06 | 98.06 | 0.99 | **98** | 1.94 | 9.641 |
| Fine KNN | 97.9 | 97.86 | 97.88 | 0.98 | 97.9 | 2.1 | 9.8311 |
| CSVM | 97.26 | 97.23 | 97.24 | 1 | 97.2 | 2.74 | 7.0433 |
| MGSVM | 96.93 | 96.9 | 96.91 | 0.99 | 96.9 | 3.07 | 8.4764 |
| Weighted KNN | 96.46 | 96.4 | 96.42 | 0.99 | 96.4 | 3.54 | 8.8949 |
| $WN^2$ | 96.36 | 96.3 | 96.32 | 0.99 | 96.3 | 3.64 | 9.8142 |
| QSVM | 96.16 | 96.13 | 96.14 | 0.99 | 96.1 | 3.84 | 6.8105 |
| $MN^2$ | 95.73 | 95.7 | 95.71 | 0.99 | 95.7 | 4.27 | **4.1722** |
| $BiN^2$ | 95.5 | 95.43 | 95.46 | 0.98 | 95.4 | 4.5 | 12.458 |
| $TriN^2$ | 95.1 | 95.03 | 95.06 | 0.98 | 95 | 4.9 | 21.856 |

*Note*: Bold denotes the best results.

Abbreviations: BiN2, bilayered neural network; CSVM, cubic SVM; ESKNN, ensembled subspace KNN; MGSVM, medium Gaussian SVM; MN2, medium neural network; TriN2, trilayered neural network; WN2, wide neural network.

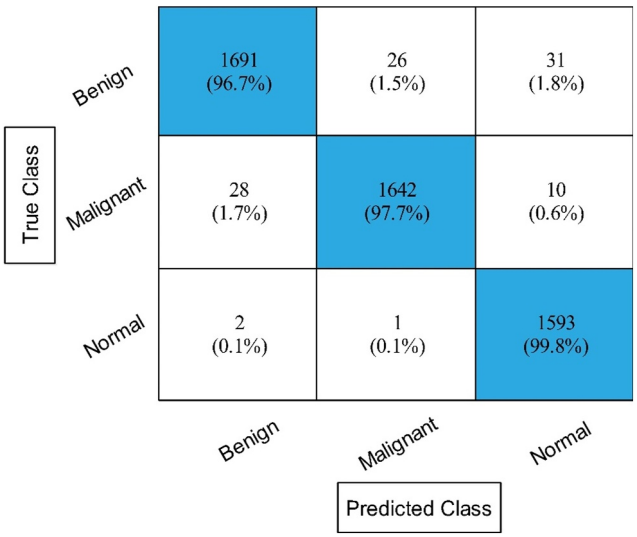


**FIGURE 9** Experiment no 3 confusion matrix for ESKNN classifier. ESKNN, Ensembled Subspace KNN.

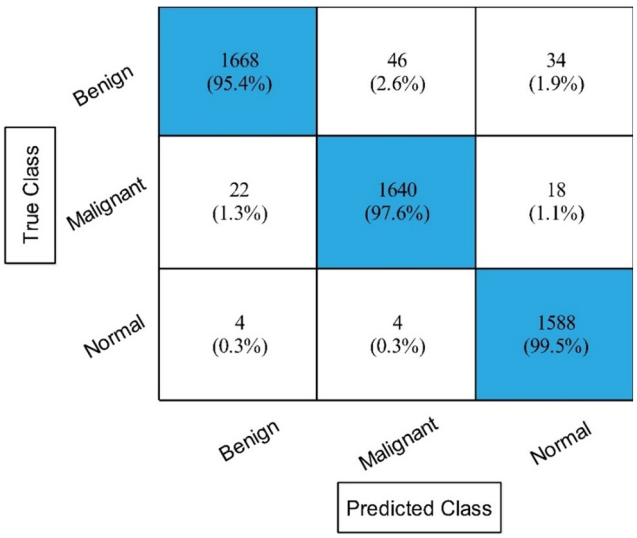


**FIGURE 10** ESKNN classifier confusion matrix for experiment no 4. ESKNN, Ensembled Subspace KNN.

Figure 10. From this figure, the diagonal values show the true prediction rate. The second-best accuracy is obtained by the Fine KNN classifier of 97%. The minimum obtained accuracy for this experiment is 90.7% by Tri-layered NN. The computational time of each classifier is also noted, and it is observed that the medium NN has a minimum time of 9.0573 (sec). Compared to the prior experiment, this one's accuracy is minor but significantly reduced in the computational time.

In the fifth experiment, selected features of both deep models are fused using proposed parallel zero-padding correlation coefficient approach. Table 5 presents the numerical results of this experiment. In this table, it is noted that the ESKNN classifier obtained the highest accuracy of 98.4%. The SER of this classifier is 98.43%, PREC is 98.36%, F1-Score is 98.39%, and FNR is 1.57%. These measures can be verified by a confusion matrix, as illustrated in Figure 11. From this figure, the diagonal values show the true prediction rate. The second best accuracy is obtained by Fine KNN classifier of 97.7%. The minimum obtained accuracy for this experiment is 96% by Bi-layered NN. Computational time of each classifier is also noted and it is observed that the medium NN has the minimum time of 9.4891 (sec). In this experiment, the accuracy is increased that is strength of this approach but on the other side, the computational time is little increased that is a drawback. Therefore, we employed again feature selection technique and tried to reduce the redundant information.

As mentioned in the previous experiment, the fusion process increased the computational time; therefore, we employed again improved feature selection technique. Results after employing feature selection are given in Table 6. The ESKNN obtained the highest accuracy of 98%. The SER of this classifier is 98.06%, PREC is 97.96%, F1-Score is 98.01%,

**TABLE 4** Classification results of EfficientNet-b0 deep model after employing improved cuckoo search based best feature selection on augmented BUSI dataset.

| Classifier | SER (%) | PREC (%) | F1S | AUC | Acc (%) | FNR | Time (sec) |
|---|---|---|---|---|---|---|---|
| ESKNN | 97.5 | 97.43 | 97.46 | 0.99 | **97.5** | 2.5 | 25.96 |
| Fine KNN | 97.06 | 97.03 | 97.04 | 0.98 | 97 | 2.94 | 25.824 |
| CSVM | 95.46 | 95.4 | 95.42 | 0.99 | 95.4 | 4.54 | 30.524 |
| MGSVM | 94.83 | 94.76 | 94.79 | 0.99 | 94.7 | 5.17 | 42.192 |
| Weighted KNN | 94.16 | 94.06 | 94.10 | 0.99 | 94 | 5.84 | 25.314 |
| $WN^2$ | 93.8 | 93.76 | 93.78 | 0.98 | 93.7 | 6.2 | 12.171 |
| QSVM | 93.8 | 93.7 | 93.74 | 0.99 | 93.7 | 6.2 | 34.042 |
| $MN^2$ | 93 | 93 | 93 | 0.98 | 92.9 | 7 | **9.0573** |
| $BiN^2$ | 90.9 | 90.86 | 90.88 | 0.95 | 90.8 | 9.1 | 33.134 |
| $TriN^2$ | 90.83 | 90.8 | 90.81 | 0.95 | 90.7 | 9.17 | 39.728 |

*Note*: Bold denotes the best results.

Abbreviations: BiN2, bilayered neural network; CSVM, cubic SVM; ESKNN, ensembled subspace KNN; MGSVM, medium Gaussian SVM; MN2, medium neural network; TriN2, trilayered neural network; WN2, wide neural network.

**TABLE 5** Classification results of proposed parallel based zero-padding correlation coefficient feature fusion.

| Classifier | SER (%) | PREC (%) | F1S | AUC | Acc (%) | FNR | Time (sec) |
|---|---|---|---|---|---|---|---|
| ESKNN | 98.43 | 98.36 | 98.39 | 0.99 | **98.4** | 1.57 | 40.95 |
| Fine KNN | 97.70 | 97.73 | 97.71 | 0.98 | 97.7 | 2.3 | 34.785 |
| CSVM | 98.13 | 98.13 | 98.13 | 1 | 98.1 | 1.87 | 33.213 |
| MGSVM | 98.00 | 97.90 | 97.94 | 1 | 97.9 | 2.0 | 49.666 |
| Weighted KNN | 96.70 | 96.60 | 96.64 | 1 | 96.6 | 3.3 | 34.658 |
| $WN^2$ | 97.30 | 97.30 | 97.30 | 1 | 97.3 | 2.7 | 11.973 |
| QSVM | 97.80 | 97.80 | 97.80 | 1 | 97.8 | 2.2 | 33.895 |
| $MN^2$ | 97.10 | 97.06 | 97.08 | 1 | 97.1 | 2.9 | **9.4891** |
| $BiN^2$ | 96.03 | 96.00 | 96.01 | 0.98 | 96 | 3.97 | 13.414 |
| $TriN^2$ | 96.16 | 96.13 | 96.14 | 0.99 | 96.1 | 3.84 | 25.693 |

*Note*: Bold denotes the best results.

Abbreviations: BiN2, bilayered neural network; CSVM, cubic SVM; ESKNN, ensembled subspace KNN; MGSVM, medium Gaussian SVM; MN2, medium neural network; TriN2, trilayered neural network; WN2, wide neural network.

and FNR is 1.94%. Confusion matrix of this classifier is also illustrated in Figure 12 that can be utilised to verify these computed measures. Computational time of each classifier is also noted after employing feature selection and it is observed that the medium NN has the minimum time of 4.9363 (sec). Comparatively to the previous experiment, this one significantly reduced the computational time.

## 5.1 | Discussion and analysis

A brief discussion of this work has been added in this section. Initially, we analyse the performance of selected pre-trained models by using Grad-CAM based visualisation. In the visualisation process, the trained models are predicted and generated heatmap as illustrated in Figure 13. In this figure, it is shown that the original images are predicted at the initial stage by using pre-trained model features and Softmax classifier. After that, the Grad-CAM based heatmap is generated on the predicted images that show the important regions (pixels) for the classification.

In the next step, we compared the proposed framework middle steps importance based on the computation time. Figure 14 showing the time comparison among all middle steps. In this figure, it is shown that the computation time of the initial modified deep learning models such as R-18 and E-B0 has been higher side, so we employed the optimisation algorithm and time is almost 50% reduced. To further maintain the accuracy, we performed the fusion and then applied optimisation technique. Hence, after employing optimisation technique, time is significantly reduced and accuracy is sufficient to compare with recent works.

In the last, a brief comparison of the proposed framework is conducted with some recent techniques in terms of accuracy. Table 7 presented a brief comparison with recent methods and it is show that the proposed accuracy is improved than the previous methods. The main strength of the proposed work is achieved improved accuracy and in very short computation time and the optimisation of extracted features compact some essential information.

## 6 | CONCLUSIONS

A new framework is proposed in this work for breast cancer classification from BUS images using deep learning optimal feature fusion and Grad-CAM visualisation. We developed a deep pipeline for the correct classification of breast lesion using ultrasound images in very precise time. Data augmentation is performed earlier to increase diversity of samples and to get better generalisation during training the selected models. The selected models are modified by adding GRU and multi head self-attention activations. Then, the modified models are trained using a deep transfer learning process and analysed through Grad-CAM visualisation. The performance of both modified models was insufficient; therefore, we developed an improved feature selection technique that selects the best features and shows an improvement in accuracy. However, the obtained accuracy was insufficient; therefore, a new fusion technique was employed to increase the classification accuracy. This step's drawback was increased computational time; therefore, we again employed a feature selection technique in

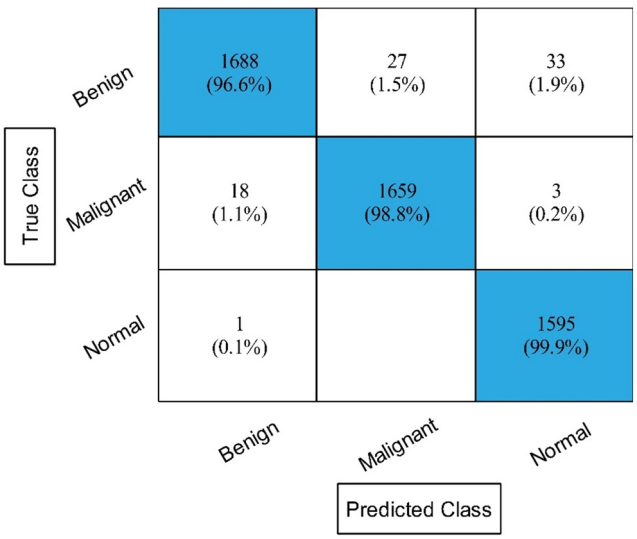


**FIGURE 11** The confusion matrix of ESKNN after employing proposed feature fusion approach. ESKNN, Ensembled Subspace KNN.

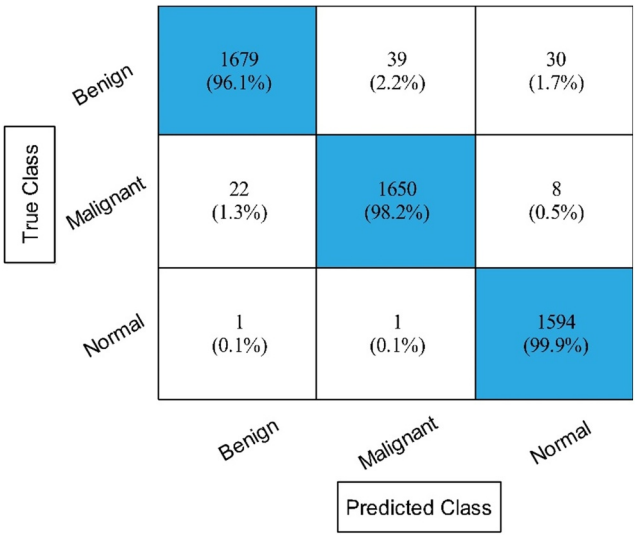


**FIGURE 12** The confusion matrix of ESKNN after employing improved feature selection technique on the fused feature vector. ESKNN, Ensembled Subspace KNN.

**TABLE 6** Classification results of improved feature selection employing on fused feature vector.

| Classifier | SER (%) | PREC (%) | F1S | AUC | Acc (%) | FNR | Time (sec) |
|---|---|---|---|---|---|---|---|
| ESKNN | 98.06 | 97.96 | 98.01 | 0.99 | **98** | 1.94 | 17.83 |
| Fine KNN | 97.36 | 97.3 | 97.32 | 0.98 | 97.3 | 2.64 | 17.059 |
| CSVM | 97.33 | 97.33 | 97.33 | 1 | 97.3 | 2.67 | 16.472 |
| MGSVM | 97.3 | 97.26 | 97.28 | 1 | 97.3 | 2.7 | 22.523 |
| Weighted KNN | 96.36 | 96.23 | 96.29 | 1 | 96.2 | 3.64 | 17.716 |
| $WN^2$ | 96.46 | 96.4 | 96.42 | 0.99 | 96.4 | 3.54 | 7.3086 |
| QSVM | 96.8 | 96.76 | 96.78 | 1 | 96.7 | 3.2 | 15.197 |
| $MN^2$ | 95.96 | 95.9 | 95.92 | 0.99 | 95.9 | 4.04 | **4.9363** |
| $BiN^2$ | 94.63 | 94.56 | 94.59 | 0.97 | 94.6 | 5.37 | 8.4681 |
| $TriN^2$ | 94.83 | 94.83 | 94.83 | 0.98 | 94.8 | 5.17 | 13.998 |

*Note*: Bold denotes the best results.

Abbreviations: BiN2, bilayered neural network; CSVM, cubic SVM; ESKNN, ensembled subspace KNN; MGSVM, medium Gaussian SVM; MN2, medium neural network; TriN2, trilayered neural network; WN2, wide neural network.

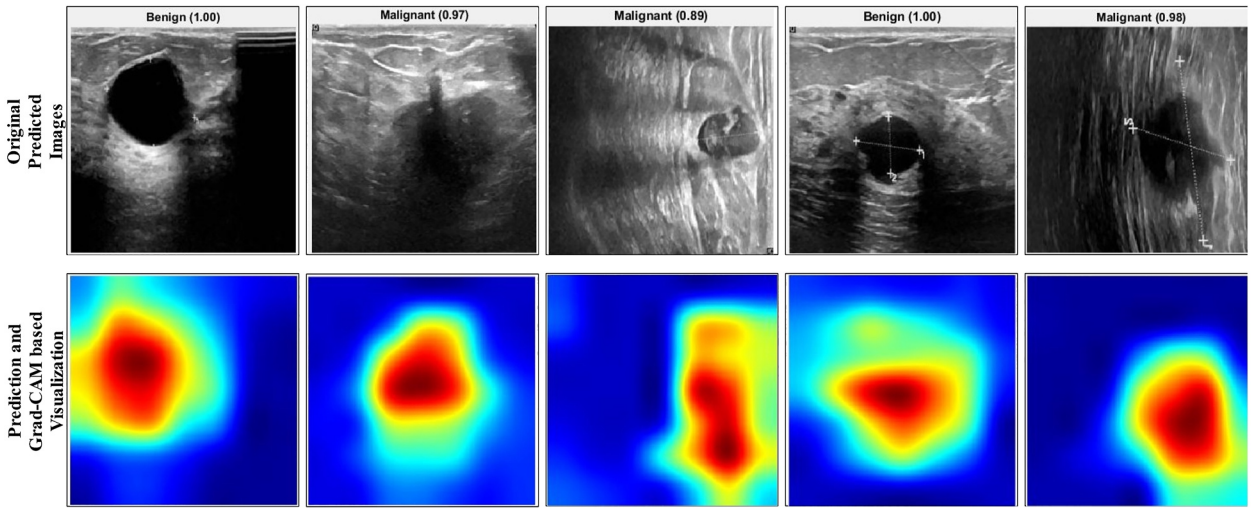


**FIGURE 13** Labelled prediction and Grad-CAM based visualisation of important region.

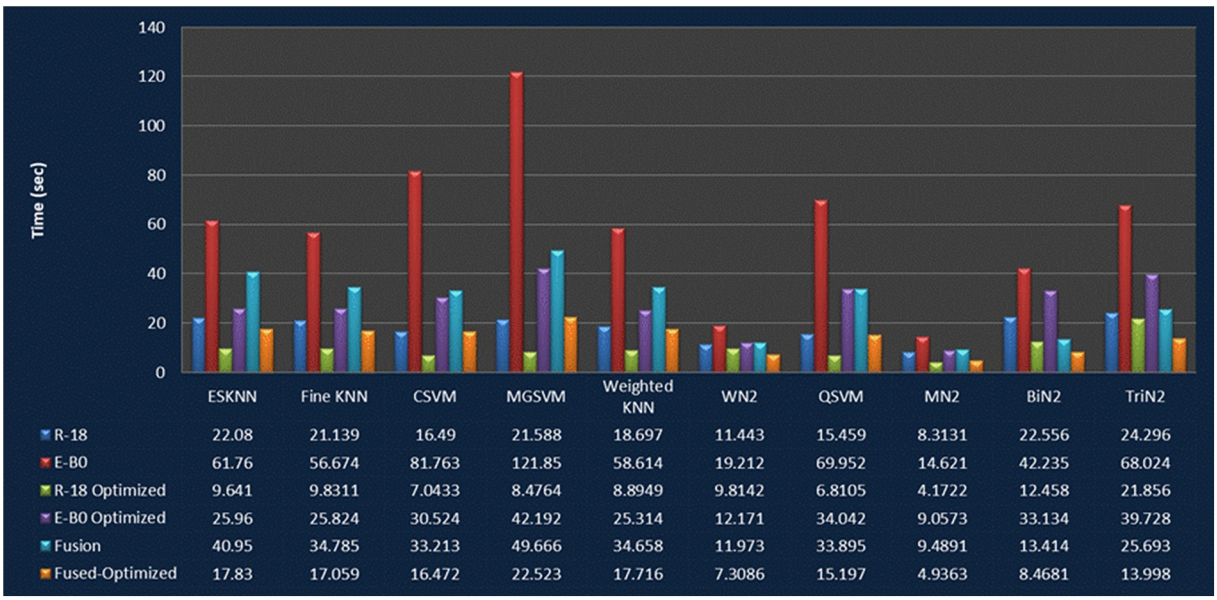


**FIGURE 14** Comparison of the proposed framework middle steps based on computation time. *R-18 denotes modified ResNet-18 model, E-B0 denotes the modified EfficientNet-B0 model with GRU module. GRU, Gated Recurrent Unit.

**TABLE 7** Comparison with recent techniques for the proposed framework.

| Reference | Year | Dataset | Accuracy (%) |
|---|---|---|---|
| [36] | 2022 | BUSI | 96.24 |
| [37] | 2022 | BUSI | 94.5 |
| [38] | 2022 | BUSI | 97.18 |
| [39] | 2022 | BUSI | 94.8 |
| [40] | 2022 | BUSI | 74.5 |
| [41] | 2022 | BUSI | 95.81 |
| **Proposed** | **2023** | **BUSI** | **98** |

*Note*: Bold denotes the best results.

Abbreviation: BUSI, breast ultrasound imaging dataset.

order to decrease the computation time. The experimental process of the proposed framework is conducted on a publicly available BUSI and obtained 98% accuracy.

## 6.1 | Limitations and future directions

The proposed framework obtained improved accuracy and executed in less computational time which is considered as an advantage of work, still there are several limitations exist, such as: (i) mixed and small-scaled datasets degrade the classification accuracy; (ii) the selection of best features reduced some important features that may be useful for the better weights visualisation and accurate classification. In the future, we will

consider more diverse datasets for the experimental process and resolve the imbalance problem. In addition, a lightweight multi-scale framework will be considered for accurate classification.

## ACKNOWLEDGEMENTS

This work is supported by the Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2024R506), the Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

## CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

## DATA AVAILABILITY STATEMENT

The dataset used in this work is publicly available: https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset.

## ORCID

*Muhammad Attique Khan* https://orcid.org/0000-0002-6347-4890
*Isaac Ofori* https://orcid.org/0000-0002-1101-0613

## REFERENCES

1. Guo, Y., Şengür, A., Tian, J.-W.: A novel breast ultrasound image segmentation algorithm based on neutrosophic similarity score and level set. Comput. Methods Progr. Biomed. 123, 43–53 (2016). https://doi.org/10.1016/j.cmpb.2015.09.007
2. Zhang, G., et al.: SHA-MTL: soft and hard attention multi-task learning for automated breast cancer ultrasound image segmentation and classification. Int. J. Comput. Assist. Radiol. Surg. 16(10), 1719–1725 (2021). https://doi.org/10.1007/s11548-021-02445-7
3. Rakha, E., Ellis, I.O.: An overview of assessment of prognostic and predictive factors in breast cancer needle core biopsy specimens. J. Clin. Pathol. 60(12), 1300–1306 (2007). https://doi.org/10.1136/jcp.2006.045377
4. Doi, K.: Computer-aided diagnosis in medical imaging: historical review, current status and future potential. Comput. Med. Imag. Graph. 31(4-5), 198–211 (2007). https://doi.org/10.1016/j.compmedimag.2007.02.002
5. Yu, K., Chen, S., Chen, Y.: Tumor segmentation in breast ultrasound image by means of res path combined with dense connection neural network. Diagnostics 11(9), 1565 (2021). https://doi.org/10.3390/diagnostics11091565
6. Luo, Y., Huang, Q., Li, X.: Segmentation information with attention integration for classification of breast tumor in ultrasound image. Pattern Recogn. 124, 108427 (2022). https://doi.org/10.1016/j.patcog.2021.108427
7. Ilesanmi, A.E., Chaumrattanakul, U., Makhanov, S.S.: Methods for the segmentation and classification of breast ultrasound images: a review. J. Ultrasound 24(4), 367–382 (2021). https://doi.org/10.1007/s40477-020-00557-5
8. Huang, K., et al.: Semantic segmentation of breast ultrasound image with fuzzy deep learning network and breast anatomy constraints. Neurocomputing 450, 319–335 (2021). https://doi.org/10.1016/j.neucom.2021.04.012
9. Ramamoorthy, S.A., Gopi, V.P.: Breast ultrasound image processing. In: Research Anthology on Medical Informatics in Breast and Cervical Cancer, pp. 586–609. IGI Global (2023)
10. Goudarzi, S., et al.: Segmentation of arm ultrasound images in breast cancer-related lymphedema: a database and deep learning algorithm. IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng. 70(9), 2552–2563 (2023). https://doi.org/10.1109/tbme.2023.3253646
11. Wang, K., et al.: Breast ultrasound image segmentation: a coarse-to-fine fusion convolutional neural network. Med. Phys. 48(8), 4262–4278 (2021). https://doi.org/10.1002/mp.15006
12. Mohammed, A., et al.: Breast cancer detection using deep learning technique based on ultrasound image. arXiv preprint arXiv:2312.05261 (2023)
13. Zhuang, Z., et al.: Breast ultrasound tumor image classification using image decomposition and fusion based on adaptive multi-model spatial feature fusion. Comput. Methods Progr. Biomed. 208, 106221 (2021). https://doi.org/10.1016/j.cmpb.2021.106221
14. Daoud, M.I., et al.: A fusion-based approach for breast ultrasound image classification using multiple-ROI texture and morphological analyses. Comput. Math. Methods Med. 2016, 1–12 (2016). https://doi.org/10.1155/2016/6740956
15. Nasiri, S., Zenkert, J., Fathi, M.: A medical case-based reasoning approach using image classification and text information for recommendation. In: Advances in Computational Intelligence: 13th International Work-Conference on Artificial Neural Networks, IWANN 2015, Palma de Mallorca, Spain, June 10-12, 2015. Proceedings, Part II, vol. 13, pp. 43–55 (2015). https://doi.org/10.1007/978-3-319-19222-2_4
16. Khosravani, M.R., Nasiri, S., Weinberg, K.: Prediction of fracture in sandwich-structured composite joints using case-based reasoning approach. Procedia Struct. Integr. 13, 168–173 (2018). https://doi.org/10.1016/j.prostr.2018.12.028
17. Caruccio, L., et al.: Can ChatGPT provide intelligent diagnoses? A comparative study between predictive models and ChatGPT to define a new medical diagnostic bot. Expert Syst. Appl. 235, 121186 (2024). https://doi.org/10.1016/j.eswa.2023.121186
18. Zhou, S., et al.: Shearlet-based texture feature extraction for classification of breast tumor in ultrasound image. Biomed. Signal Process Control 8(6), 688–696 (2013). https://doi.org/10.1016/j.bspc.2013.06.011
19. Hossain, S., et al.: Vision transformers, ensemble model, and transfer learning leveraging explainable ai for brain tumor detection and classification. IEEE J. Biomed. Health Inform. 28(3), 1261–1272 (2023). https://doi.org/10.1109/jbhi.2023.3266614
20. Daoud, M.I., Abdel-Rahman, S., Alazrai, R.: Breast ultrasound image classification using a pre-trained convolutional neural network. In: 2019 15th International Conference on Signal-Image Technology and Internet-Based Systems (SITIS), pp. 167–171 (2019)
21. Ding, J., et al.: Breast ultrasound image classification based on multiple-instance learning. J. Digit. Imag. 25(5), 620–627 (2012). https://doi.org/10.1007/s10278-012-9499-x
22. Elsborg, J., Salvatore, M.: Using LLM models and explainable ML to analyse biomarkers at single cell level for improved understanding of diseases. bioRxiv 08(24), 554441 (2023)
23. Singh, V.K., et al.: An efficient solution for breast tumor segmentation and classification in ultrasound images using deep adversarial learning. arXiv preprint arXiv:1907.00887 (2019)
24. Yusufiyah, H.K.N., et al.: Feature extraction for classifying lesion's shape of breast ultrasound images. In: 2015 2nd International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE), pp. 102–106 (2015)
25. Shi, J., et al.: Stacked deep polynomial network based representation learning for tumor classification with small ultrasound image dataset. Neurocomputing 194, 87–94 (2016). https://doi.org/10.1016/j.neucom.2016.01.074
26. Bing, L., Wang, W.: Sparse representation based multi-instance learning for breast ultrasound image classification. Comput. Math. Methods Med. 2017, 1–10 (2017). https://doi.org/10.1155/2017/7894705
27. Wang, S., et al.: Explainable AI for B5G/6G: Technical Aspects, Use Cases, and Research Challenges. arXiv e-prints (2021). arXiv: 2112.04698
28. Kalafi, E.Y., et al.: Classification of breast cancer lesions in ultrasound images by using attention layer and loss ensemble in deep convolutional neural networks. Diagnostics 11(10), 1859 (2021). https://doi.org/10.3390/diagnostics11101859
29. Eroğlu, Y., Yildirim, M., Çinar, A.: Convolutional Neural Networks based classification of breast ultrasonography images by hybrid method with respect to benign, malignant, and normal using mRMR. Comput. Biol.

Med. 133, 104407 (2021). https://doi.org/10.1016/j.compbiomed.2021.104407
30. Silva, K., Frommholz, I.: What if ChatGPT wrote the abstract? explainable multi-authorship attribution with a data augmentation strategy
31. Moon, W.K., et al.: Computer-aided diagnosis of breast ultrasound images using ensemble learning from convolutional neural networks. Comput. Methods Progr. Biomed. 190, 105361 (2020). https://doi.org/10.1016/j.cmpb.2020.105361
32. Mikołajczyk, A., Grochowski, M.: Data augmentation for improving deep learning in image classification problem. In: 2018 International Interdisciplinary PhD Workshop (IIPhDW), pp. 117–122 (2018)
33. Gupta, S., et al.: A novel transfer learning-based model for ultrasound breast cancer image classification. In: Computational Vision and Bio-Inspired Computing: Proceedings of ICCVBIC 2022, pp. 511–523. Springer (2023)
34. Jabeen, K., et al.: BC2NetRF: breast cancer classification from mammogram images using enhanced deep learning features and equilibrium-jaya controlled regula falsi-based features selection. Diagnostics 13(7), 1238 (2023). https://doi.org/10.3390/diagnostics13071238
35. Pramanik, P., et al.: Deep feature selection using local search embedded social ski-driver optimization algorithm for breast cancer detection in mammograms. Neural Comput. Appl. 35(7), 5479–5499 (2023). https://doi.org/10.1007/s00521-022-07895-x
36. Mishra, A.K., et al.: Achieving highly efficient breast ultrasound tumor classification with deep convolutional neural networks. Int. J. Inf. Technol. 14(7), 1–10 (2022). https://doi.org/10.1007/s41870-022-00901-4
37. Cui, W., et al.: FMRNet: a fused network of multiple tumoral regions for breast tumor classification with ultrasound images. Med. Phys. 49(1), 144–157 (2022). https://doi.org/10.1002/mp.15341
38. Liu, H., et al.: Artificial intelligence-based breast cancer diagnosis using ultrasound images and grid-based deep feature generator. Int. J. Gen. Med. 15, 2271–2282 (2022). https://doi.org/10.2147/ijgm.s347491
39. Zeng, W., et al.: Identification of maize leaf diseases by using the SKPSNet-50 convolutional neural network model. Sustainable Computing: Informatics and Systems 35, 100695 (2022). https://doi.org/10.1016/j.suscom.2022.100695
40. Shen, W.-Q., et al.: Using an improved residual network to identify PIK3CA mutation status in breast cancer on ultrasound image. Front. Oncol. 12 (2022). https://doi.org/10.3389/fonc.2022.850515
41. Yan, Y., et al.: Accurate segmentation of breast tumors using AE U-net with HDC model in ultrasound images. Biomed. Signal Process Control 72, 103299 (2022). https://doi.org/10.1016/j.bspc.2021.103299

**How to cite this article:** Jabeen, K., et al.: An EfficientNet integrated ResNet deep network and explainable AI for breast lesion classification from ultrasound images. CAAI Trans. Intell. Technol. 10(3), 842–857 (2025). https://doi.org/10.1049/cit2.12385